– Mobile phones are strictly prohibited in the examination hall.
– Scientific calculators are not permitted during the exam.
– Answers written with a red pen or pencil will not be marked.
– Do not divide the page into two columns.

Exercise 1 (1.5 + 1 + 1 points). Consider the sequence defined recursively by

$$x_1 = \sqrt{2}, \quad x_{n+1} = \sqrt{2 + x_n}.$$

1. Show, that $x_n < 2$, for all $n \in \mathbb{N}$.
2. Show, that $x_n < x_{n+1}$, for all $n \in \mathbb{N}$.
3. Find the limit of (x_n).

Exercise 2 (1.5 + 2 points)

1. Let $f : \mathbb{R} \longrightarrow \mathbb{R}$ be a periodic function with period T. Show that, $\forall n \in \mathbb{N}, nT$ is a period too.
2. Let $f : \mathbb{R} \longrightarrow \mathbb{R}$ be a function such that $\forall x \in \mathbb{R} : f(x) \neq 0$, and $\quad f(x+1) = \frac{f(x)-5}{f(x)-3}$. Show that $T = 4$, is a period for f.

Exercise 3 (1.5 + 1.5 + 0.5 + 2 points).

1. Let $a_1, a_2, ..., a_n$ be real positive numbers. Show that $\sum_{k=1}^{n} a_k \ln(\frac{1}{n}\sum_{k=1}^{n} a_k) \leqslant \sum_{k=1}^{n} a_k \ln(a_k)$.
2. Let $a_1, a_2, ..., a_n$ be real positive numbers. Show that $\frac{a_1}{a_2} + \frac{a_2}{a_3} + ... + \frac{a_{n-1}}{a_n} + \frac{a_n}{a_1} \geq n$.
3. Compare $[\sqrt{x}]$ with $[\sqrt{[x]}]$, where x is a nonnegative real number.
4. Let Ω be a part of $\mathbb{R}$, defined by $\Omega = \{\frac{\cos m\pi}{m} + e^{-n}, \quad n \in \mathbb{N} \cup \{0\}, m \in \mathbb{Z}^*\}$.
   (a) Show that Ω is bounded.
   (b) Determine with justification its extremums.

Exercise 4 (1.25 +1.25 + 0.5+1+1+0.5 points).
Let $k \in (0, 1)$. Consider the function f defined from $\mathbb{R}$ to $\mathbb{R}$ by

$$\begin{cases} f(x) = x^2 \sin \dfrac{1}{x} + kx, x \neq 0, \\ f(0) = 0. \end{cases}$$

1. Show that f is continuous on $\mathbb{R}$.
2. Show that f is differentiable on $\mathbb{R}$.
3. Write the expression of the derivative f' on $\mathbb{R}$.
4. Show that for all $\delta > 0$ there exist $x_1, x_2 \in (-\delta, \delta)$ such that $f'(x_1) > 0$ and $f'(x_2) < 0$.
5. Assume that f' is continuous at 0. Show that there exists $\delta_k > 0$ satisfying: $-k < f'(x) - k < k$ for all $x \in (-\delta_k, \delta_k)$.
6. Conclude.

Final Exam Correction

Exercise 1

1. By Induction. One has $x_1 = \sqrt{2} < 2$. Suppose that $x_n < 2$, and prove that $x_{n+1} < 2$. We have:
$$x_{n+1} = \sqrt{2 + x_n} < \sqrt{2+2} < 2.$$

2. $x_0 < x_1$ because:
$$\sqrt{2} < \sqrt{2 + \sqrt{2}}.$$
Suppose that $x_n < x_{n+1} \implies x_n + 2 < x_{n+1} + 2$, which means that $x_{n+1} < x_{n+2}$.

3. (x_n) is increasing and upper bounded, thus convergent. Since $f(x) = \frac{\sqrt{2+x}}{2+x}$ is continuous on $[0, \infty]$, the limit verifies:
$$l = \sqrt{2 + l}.$$
Which admits two solutions -1, and +2. -1 is rejected.

Exercise 2

1. By induction: For $n = 1$, the statement is true (definition). For $n = 2$, we have $f(x+2T) = f(x+T) = f(x)$.
Suppose that $f(x + nT) = f(x)$, then:
$$f(x + (n+1)T) = f(x + nT + T) = f(x + nT) = f(x).$$
Therefore, $f(x + nT) = f(x)$.

2. Successively, one has:
$$f(x+4) = f(x+3+1) = \frac{f(x+3) - 5}{f(x+3) - 3} = \frac{\frac{f(x+2)-5}{f(x+2)-3} - 5}{\frac{f(x+2)-5}{f(x+2)-3} - 3} = \frac{2f(x+2) - 5}{f(x+2) - 2} = \frac{2\frac{f(x+1)-5}{f(x+1)-3} - 5}{\frac{f(x+1)-5}{f(x+1)-3} - 2}$$
$$= \frac{3f(x+1) - 5}{f(x+1) - 1} = \frac{3\frac{f(x)-5}{f(x)-3} - 5}{\frac{f(x)-5}{f(x)-3} - 1} = \frac{-2f(x)}{-2} = f(x).$$

Exercise 3

1. It is sufficient to apply the generalized convex property to $f(x) = x \ln x$.

2. Use the Arithmetic Mean - Geometric Mean inequality.

3. $$[\sqrt{[x]}] \leq [\sqrt{x}].$$

4. We have
$$\Omega = \left\{ \cos\left(\frac{\pi m}{m}\right) + e^{-n} : n \in \mathbb{N}, m \in \mathbb{Z}^* \right\} = \Omega_1 + \Omega_2.$$
$$\Omega_1 = \left\{ \cos\left(\frac{\pi m}{m}\right), m \in \mathbb{Z}^* \right\}, \quad -1 \leq \Omega_1 \leq 1.$$
$$\min \Omega_1 = \inf\ \Omega_1 = -1,\ \text{for } m = 1, \quad \max \Omega_1 = \sup\ \Omega_1 = 1,\ \text{for } m = -1$$

$$\Omega_2 = \left\{e^{-n} : n \in \mathbb{N}\right\}, \quad 0 < \Omega_2 \leq 1$$

$$\sup\,\Omega_2 = \max\,\Omega_2 = 1, \text{ for } n = 0, \quad \inf\,\Omega_2 = 0. \quad (\min\,\Omega_2 \text{ doesn't exist .})$$

$$\inf\,\Omega_2 = 0 \iff \forall \epsilon > 0, \exists n_0 \in \mathbb{N} : 0 \leqslant \exp(-n_0) < \epsilon.$$

Indeed,

$$-n_0 \ln e < \ln \epsilon \implies n_0 \geq -\ln \epsilon.$$

If $\varepsilon > 1$, then it is trivial. If $0 < \varepsilon < 1$, we use the Archimedean property to show that $[-\ln \varepsilon] + 1$ is sufficient.
Thus,

$$\sup\,\Omega = \sup\,\Omega_1 + \sup\,\Omega_2 = 1 + 1 = 2 = \max\,\Omega.$$

$$\inf\,\Omega = \inf\,\Omega_1 + \inf\,\Omega_2 = -1 + 0 = -1$$

$$-1 < \Omega \leq 2.$$

Exercise 4

Let $k \in (0,1)$. consider the function f defined from $\mathbb{R}$ to $\mathbb{R}$ by

$$\begin{cases} f(x) = x^2 \sin\dfrac{1}{x} + kx, : x \neq 0 \\ f(0) = 0. \end{cases}$$

1. The function f is continuous on $\mathbb{R}^*$ and we have

$$\lim_{x\to 0} f(x) = \lim_{x\to 0}[x^2 \sin\frac{1}{x} + kx] = 0 = f(0).$$

So, f is continuous on $\mathbb{R}$.

2. The function f is differentiable on $\mathbb{R}^*$ and we have

$$f'(x) = 2x\sin\frac{1}{x} + \cos\frac{1}{x} + k, \quad x \neq 0.$$

Now,

$$\lim_{x\to 0}\frac{f(x) - f(0)}{x - 0} = \lim_{x\to 0}\frac{x^2\sin\frac{1}{x} + kx}{x} = \lim_{x\to 0}[x\sin\frac{1}{x} + k] = k.$$

Hence, f is differentiable on $\mathbb{R}$.

3. The expression of the derivative f':

$$\begin{cases} f'(x) = 2x\sin\dfrac{1}{x} + \cos\dfrac{1}{x} + k, : x \neq 0 \\ f'(0) = k. \end{cases}$$

4. Let $\delta > 0$. There exists $n \in \mathbb{N}$ such that $x_1 = \dfrac{1}{2n\pi}, x_2 = \dfrac{1}{(2n+1)\pi} \in (-\delta, \delta)$ and we have $f'(x_1) = 1 + k > 0$ and $f'(x_2) - 1 + k < 0$.

5. Assume that f' is continuous at 0. So, for $\varepsilon = k$ there exists $\delta_0 > 0$ such that for all $x \in (-\delta_k, \delta_k)$ we have $|f'(x) - f'(0)| < k$, i.e $-k < f'(x) - k < k$.

6. We get $0 < f'(x) < 2k$ for all $x \in (-\delta_k, \delta_k)$, which contradicts 4.
Hence, f' is not continuous at 0.

One has: $\forall x \geq 0$.

1) $[x] \leq x \Rightarrow \sqrt{[x]} \leq \sqrt{x} \Rightarrow [\sqrt{[x]}] \leq [\sqrt{x}]$.

2) $[\sqrt{x}] \leq \sqrt{x} \Rightarrow [\sqrt{x}]^2 \leq x$

$\Rightarrow [[\sqrt{x}]^2] = [\sqrt{x}]^2 \leq [x]$

$\Rightarrow [\sqrt{x}] \leq \sqrt{[x]}$

$\Rightarrow [[\sqrt{x}]] = [\sqrt{x}] \leq [\sqrt{[x]}]$

(1) + (2) $\Rightarrow [\sqrt{x}] = [\sqrt{[x]}]$.